TypeSafe AI

JEV: Introduction and Usage

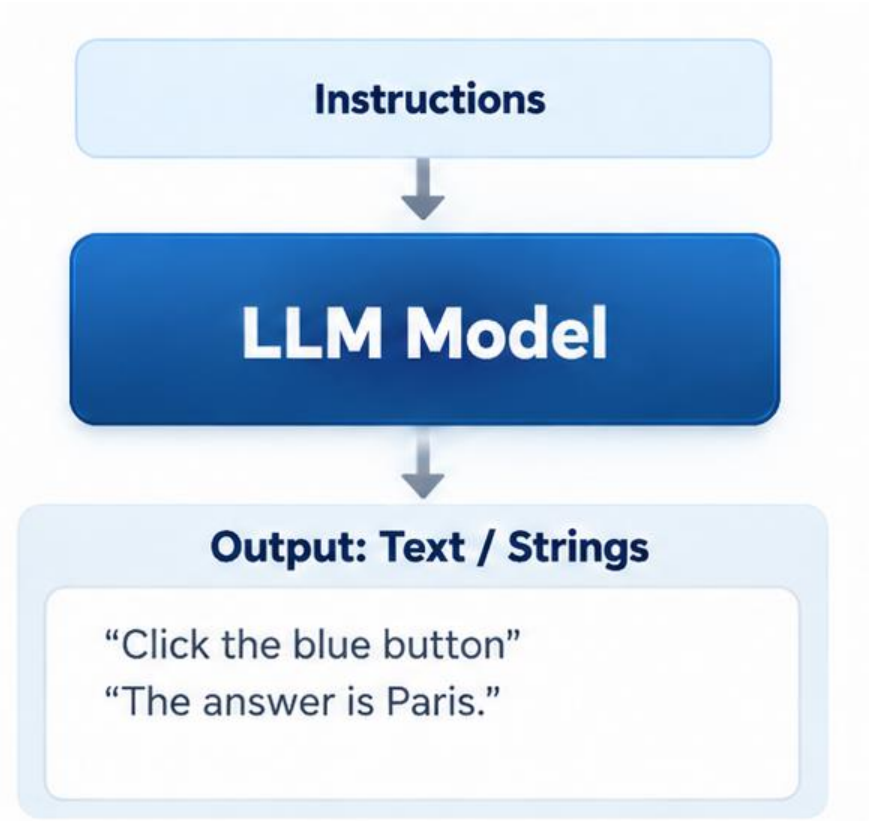
22th Oct

Shuai Wang

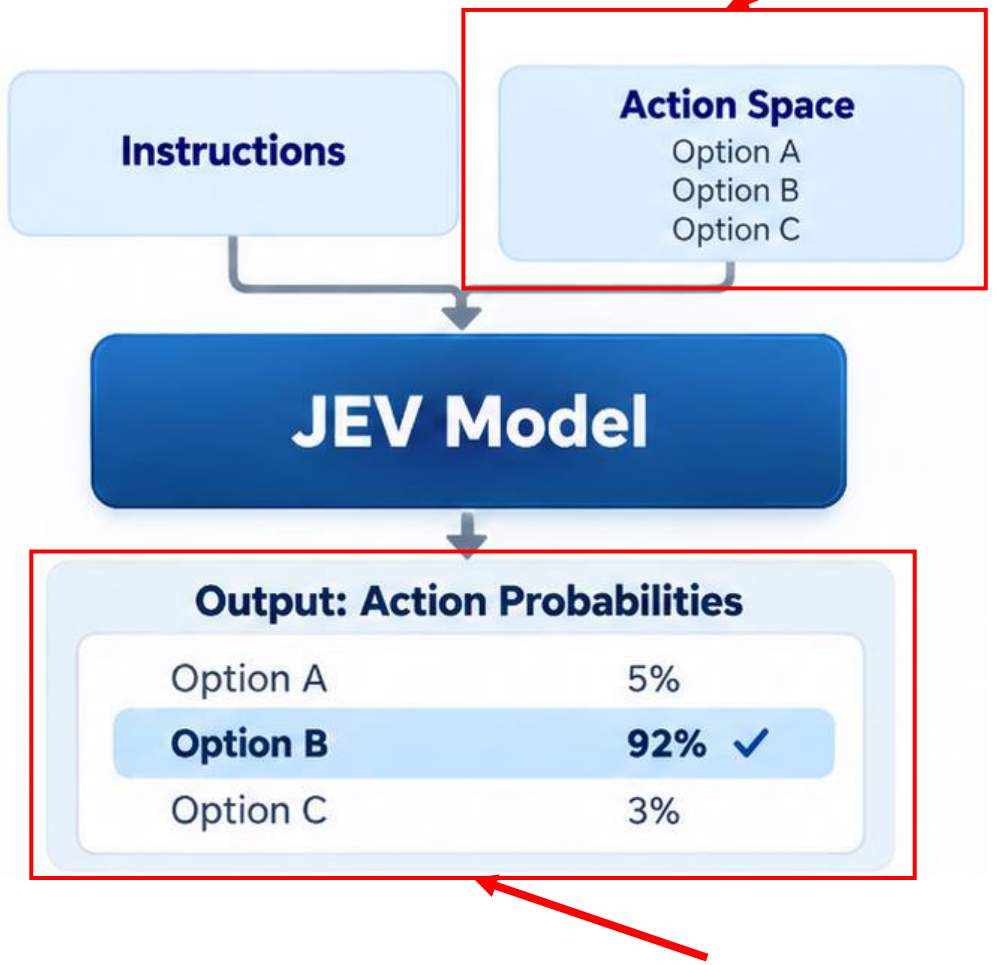
https://wangshuaiia.github.io/reading-club/

What Is a Jev Model?

➢ Traditional LLM



➢ Jev

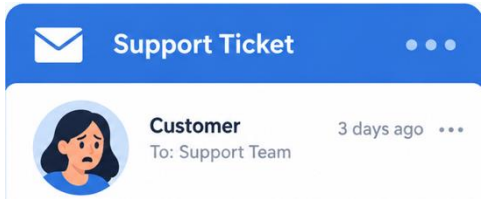


❖ Jev makes a decision instead of generating text

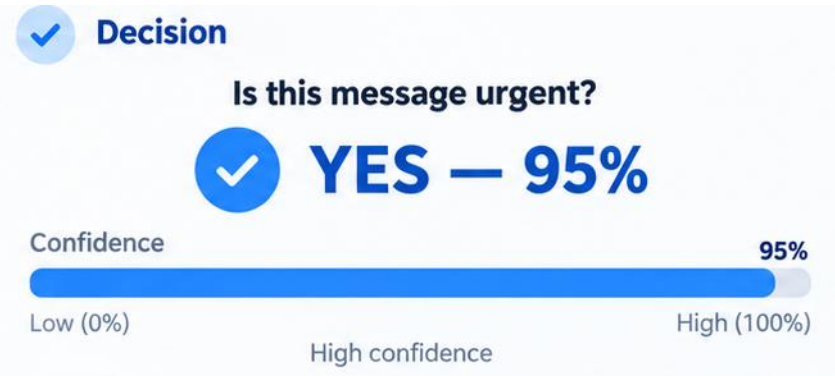
How Jev Works

➢ Example: Making a Yes/No Decision with Noul

Task: Determine whether a customer message is urgent.



"I haven't been able to connect my Stripe account for 3 days and I'm losing sales. Please help urgently!"



Input: State + Yes/No Question

```
{
  "state": "I haven't been able to connect my Stripe account for 3 days and I'm losing sales. Please help urgently!",
  "questions": {
    "is_urgent": {
      "type": "noul",
      "instructions": "Is this message urgent?"
}}}
```

Output: Probability

```
{
  "is_urgent": {
    "type": "noul",
    "noul": 0.95
  }
}
```

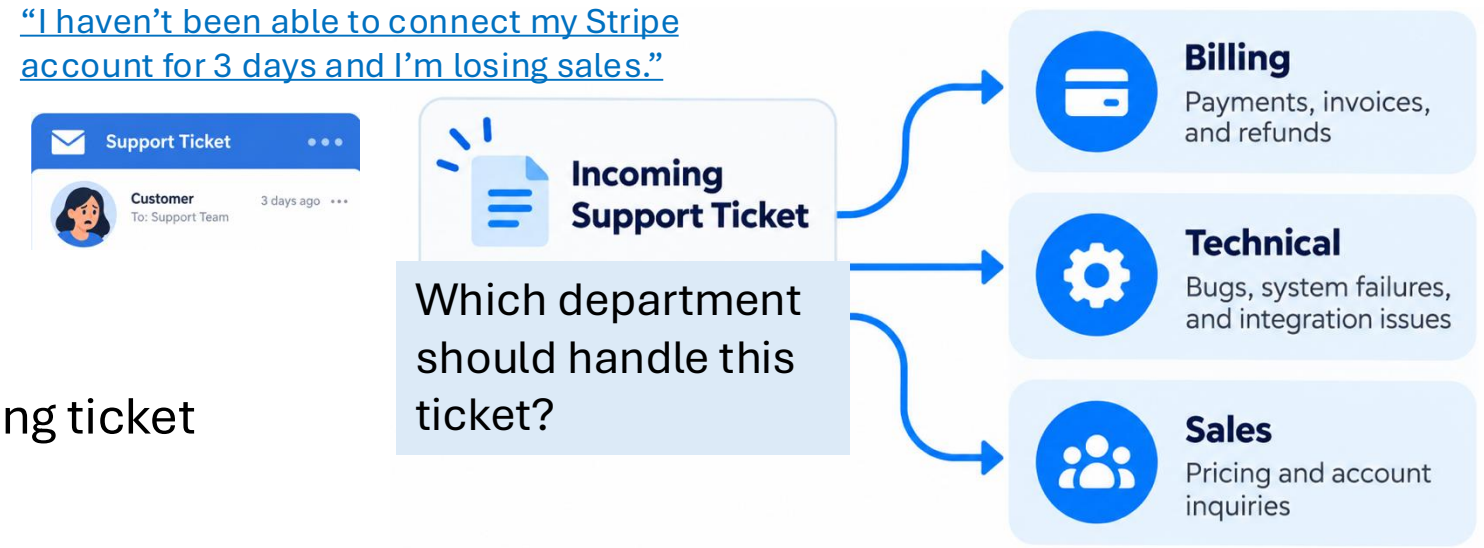
Yes -- 95%
No -- 5%

Noul directly estimates the probability of a Yes/No decision.

How Jev Works

- Example: Making a Choice with Jev

In a company, route each incoming ticket to the appropriate team.



Input: Define the decision

```
{
 "questions": {
  "department": {
   "type": "choice",
   "instructions": "Route this ticket to the appropriate team.",
   "criteria": {
      "billing": "Payments, invoices, and refunds",
      "technical": "Bugs, system failures, and integration issues",
      "sales": "Pricing and account inquiries"
}}}}
```

Output: Define + Probabilities

```
{
 "department": {
      "type": "choice",
      "choice": "billing",
      "confidence": 0.8,
      "probabilities": {
            "billing": 0.17,
            "technical": 0.83,
            "sales": 0.00}
}}
```

List the department options in criteria

Jev vs LLM in Agent Era

- LLMs are too heavy for simple decisions

  In the agent era, many steps require a decision, not text generation



…, Which tool should I use next?

"Use the browser tool next, …"

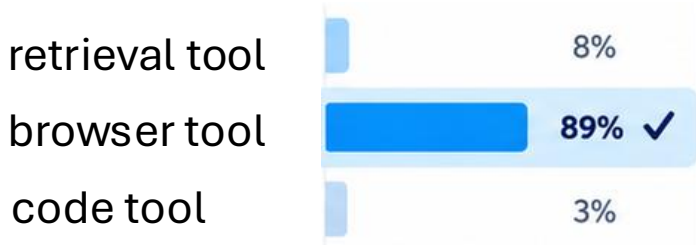
browser tool

Extra token generation
Higher latency
Parsing overhead
Expensive at scale

- Jev-based Decision

…, Which tool should I use next?



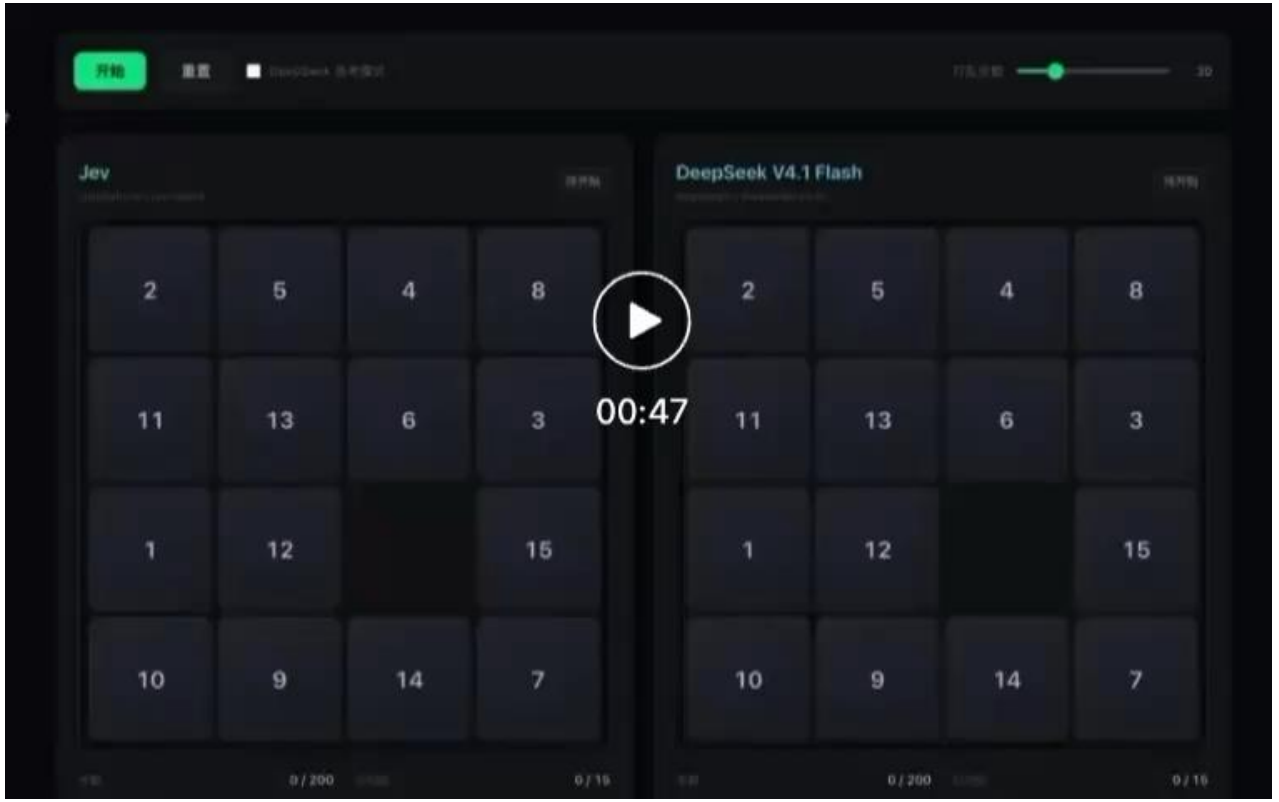


No extra token generation

Why Is JEV So Fast and Cost-Efficient?

- JEV makes decisions
  No long text generation → Faster responses + Lower cost

❖ **Speed Demo: 15-Puzzle**



Jev DeepSeek V4.1 Flash

Time:
15s VS **43s**

Time and Cost Comparison: LLM vs. JEV

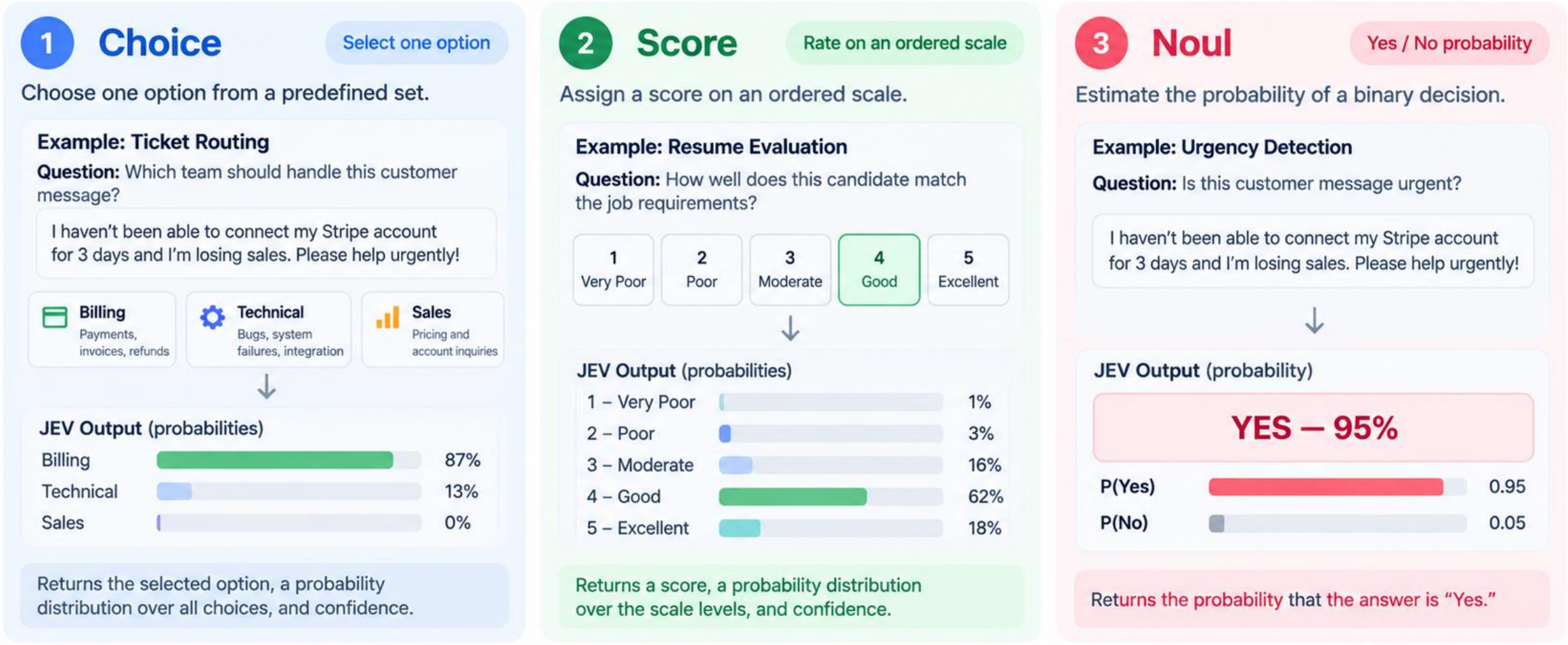
	Generative LLMs	JEV
Output	Generates text, then parses it	Returns a structured decision directly
Latency	3–329 s	70–500 ms
Input Cost	$0.20–$10 / 1M tokens	$0.042 / 1M tokens
Output Cost	Typically much higher than input	Free

❖ No autoregressive text generation ⟶ **Faster decisions + Lower cost**

How JEV Works

```
{
   "type": "noul",
    "instructions": ”Is this message urgent?"
}
```

- Three Decision Primitives

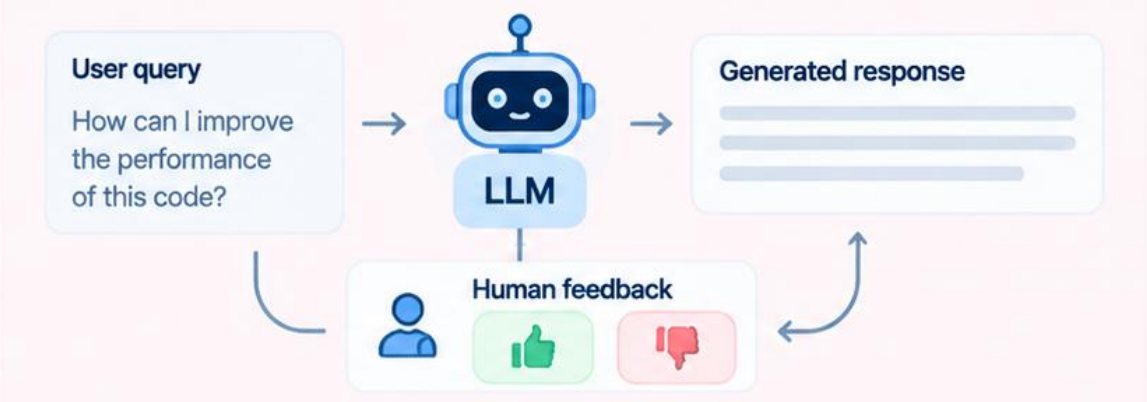


How Is JEV Trained? - RLCD

LLMs: RLHF

(Reinforcement Learning from Human Feedback)

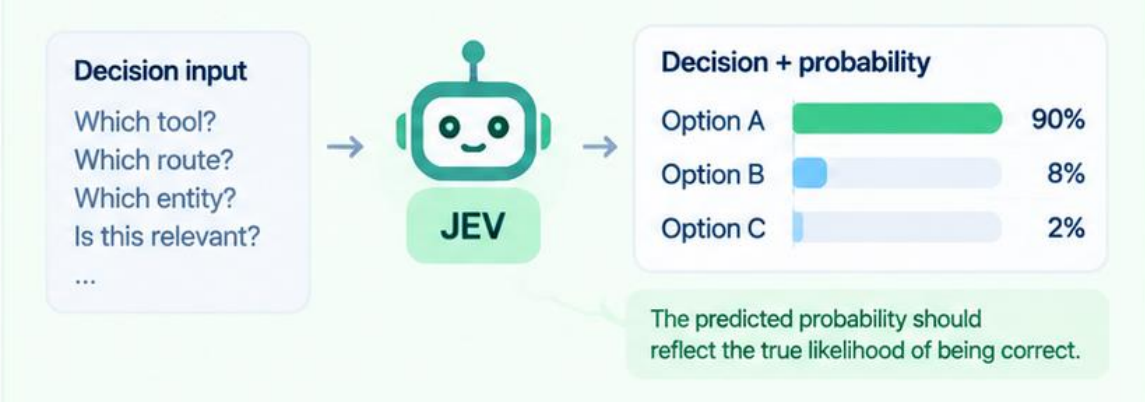
Optimize for human preference



Jev: RLCD

(Reinforcement Learning for Calibrated Decisions**)**

Optimize for calibrated decisions

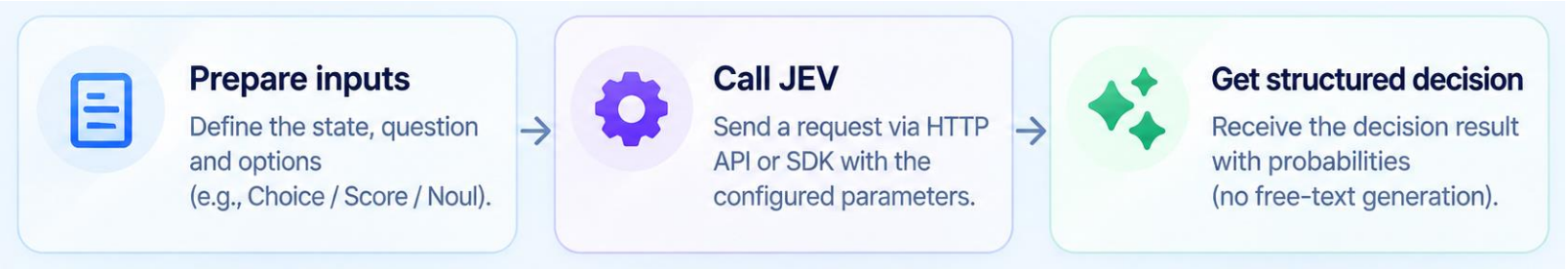


From “good-looking answers” → to “reliable decision probabilities”

How Can We Use JEV?

➢ How to access JEV

Use JEV via API / SDK

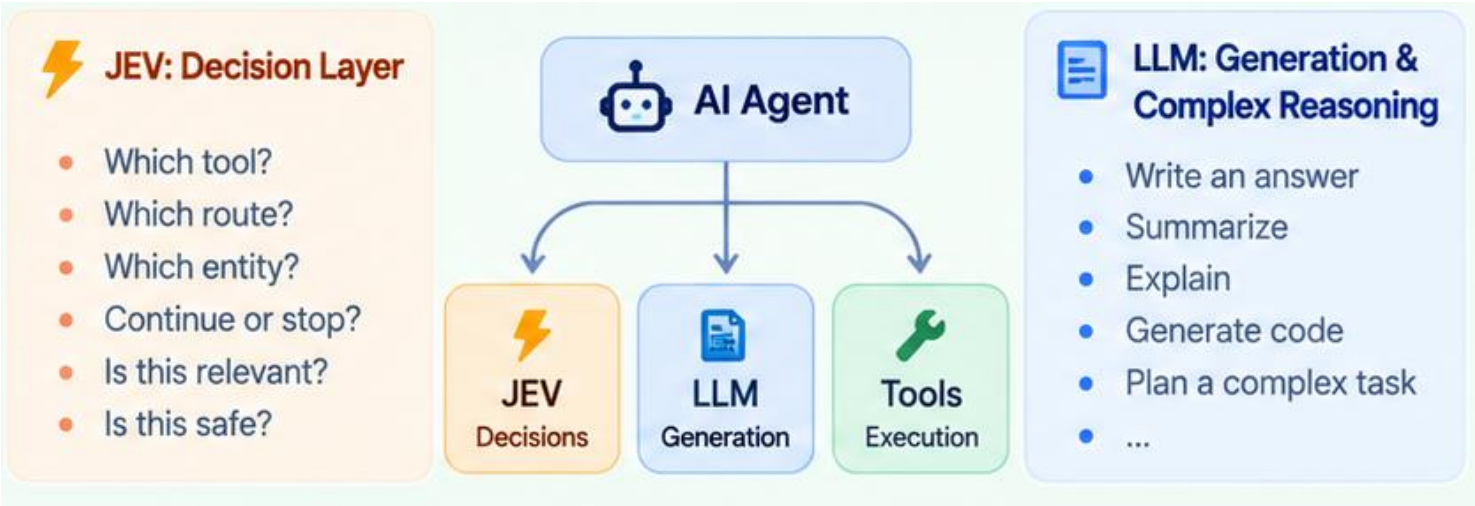


➢ Use JEV as an Agent Skill

JEV can be integrated into LLM-based agents as a **Skill**.

Official TypeSafe docs, ready for direct LLM use.

Let LLMs generate. Let JEV decide.



When Should We Use JEV?

1. Ticket & Message Classification

2. Decision Layer for AI Agents

Use JEV for repeated intermediate decisions instead of calling an LLM at every step.

Examples:

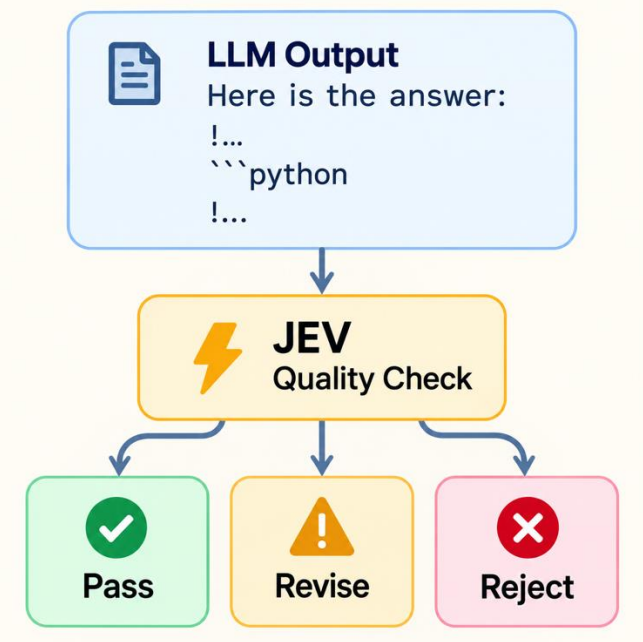
Which tool? Which route? Which entity? Continue or stop?

3. Quality Control for LLM Outputs

Make fast safety or policy decisions on incoming content.

Examples:

Is this unsafe? Is this spam? Is it correct enough?



What Can JEV Do?

4. Real-Time Game Decisions



❖ Best fit: many small decisions, repeated frequently,

Questions

- Do agents really need an LLM for every step?
- Can fast decision models replace LLM reasoning in bounded tasks?
- Could “Jev + LLM” become a better agent architecture than “LLM everywhere”?

Thank you!

Happy to take questions.